



CrysDiS

Crystal Diffraction Simulator

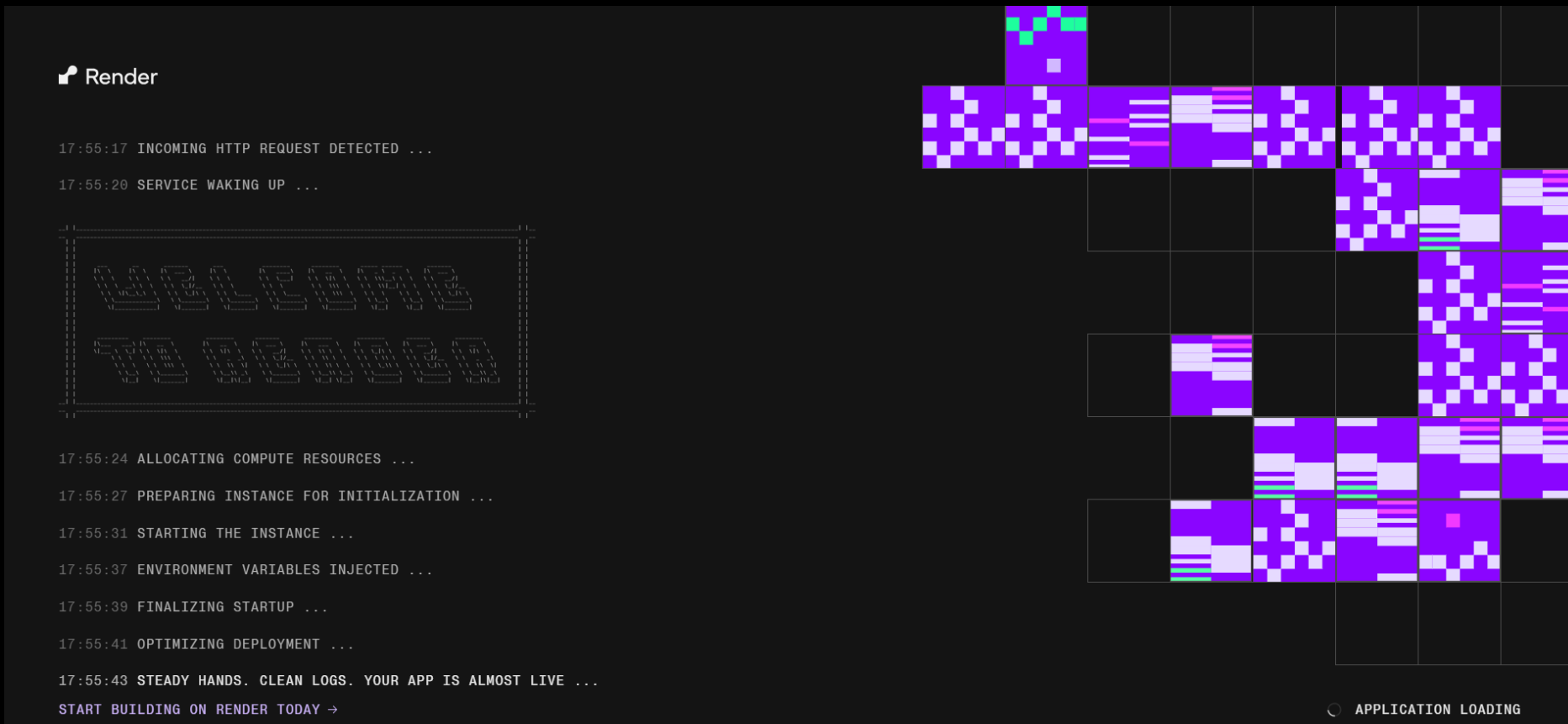
(created using ChatGPT Codex)



Start the program

Option 1: use the website version

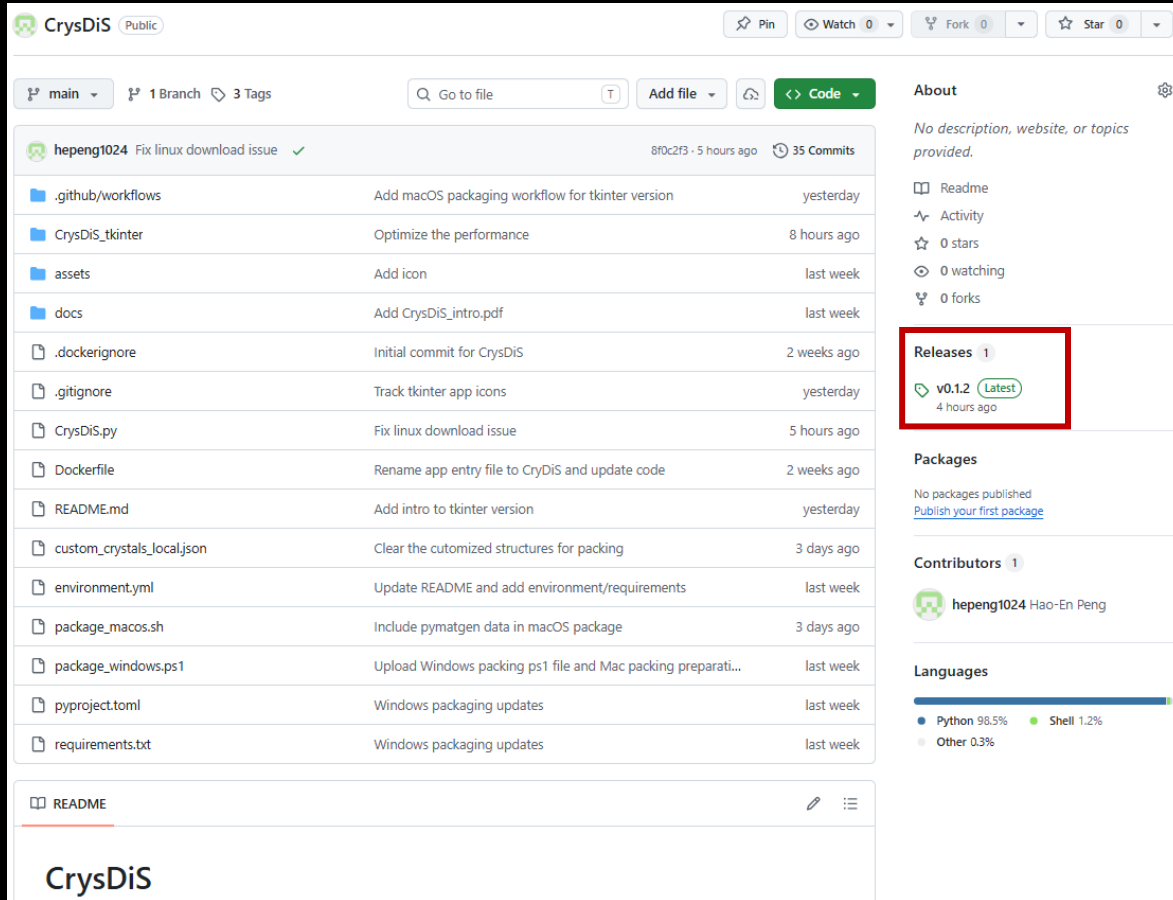
- Website: <https://crysdis.onrender.com/>
- Since this is a free URL, it may take 2~3 minutes to warm up
- It also does not have high computational performance
- If you think this software is useful, running it locally is strongly recommended



Start the program

Option 2: download the program and run it locally

- Go to <https://github.com/hepeng1024/CrysDiS> → Click “Releases” → Download the version that matches your OS



CrysDiS Public

main 1 Branch 3 Tags

Go to file Add file Code

hepeng1024 Fix linux download issue ✓ 8f0c2f3 · 5 hours ago 35 Commits

File	Commit Message	Time
.github/workflows	Add macOS packaging workflow for tkinter version	yesterday
CrysDiS_tkinter	Optimize the performance	8 hours ago
assets	Add icon	last week
docs	Add CrysDiS_intro.pdf	last week
.dockerignore	Initial commit for CrysDiS	2 weeks ago
.gitignore	Track tkinter app icons	yesterday
CrysDiS.py	Fix linux download issue	5 hours ago
Dockerfile	Rename app entry file to CryDiS and update code	2 weeks ago
README.md	Add intro to tkinter version	yesterday
custom_crystals_local.json	Clear the customized structures for packing	3 days ago
environment.yml	Update README and add environment/requirements	last week
package_macos.sh	Include pyratgen data in macOS package	3 days ago
package_windows.ps1	Upload Windows packing ps1 file and Mac packing preparati...	last week
pyproject.toml	Windows packaging updates	last week
requirements.txt	Windows packaging updates	last week

Releases 1

v0.1.2 (Latest) 4 hours ago

Assets 8

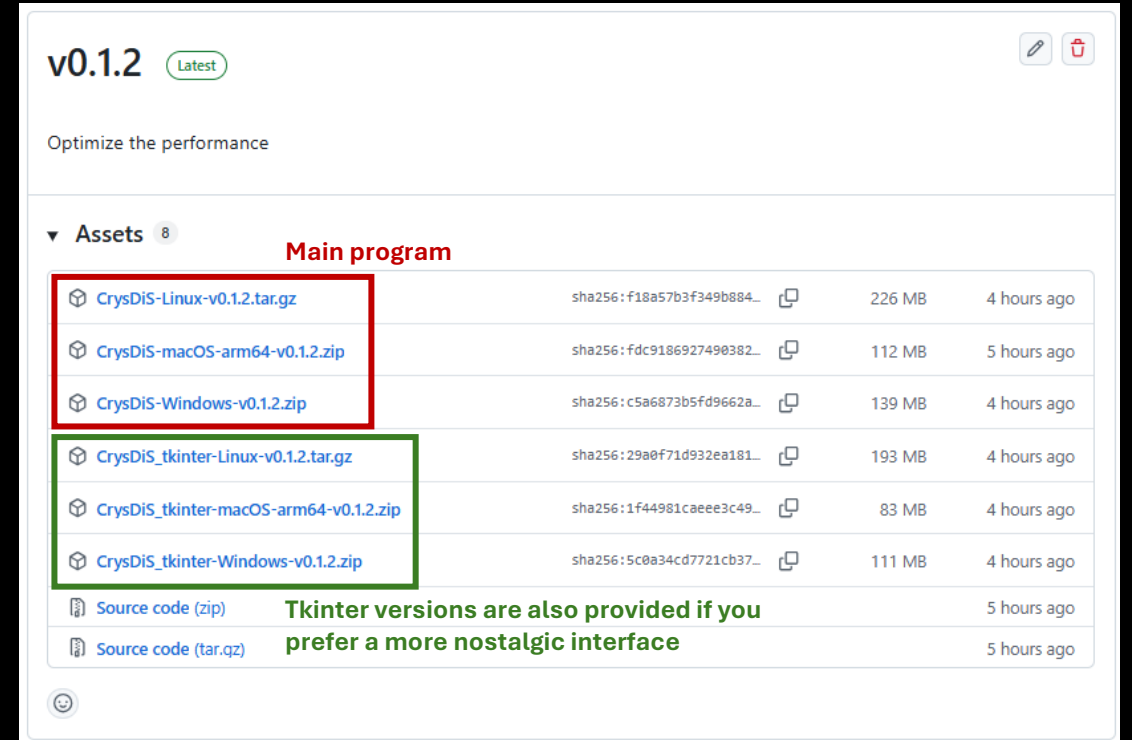
Asset	SHA256	Size	Time
CrysDiS-Linux-v0.1.2.tar.gz	sha256:f18a57b3f349b884...	226 MB	4 hours ago
CrysDiS-macOS-arm64-v0.1.2.zip	sha256:fdc9186927490382...	112 MB	5 hours ago
CrysDiS-Windows-v0.1.2.zip	sha256:c5a6873b5fd9662a...	139 MB	4 hours ago
CrysDiS_tkinter-Linux-v0.1.2.tar.gz	sha256:29a0f71d932ea181...	193 MB	4 hours ago
CrysDiS_tkinter-macOS-arm64-v0.1.2.zip	sha256:1f44981caeee3c49...	83 MB	4 hours ago
CrysDiS_tkinter-Windows-v0.1.2.zip	sha256:5c0a34cd7721cb37...	111 MB	4 hours ago
Source code (zip)			5 hours ago
Source code (tar.gz)			5 hours ago

Contributors 1

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Languages

Python 98.5% Shell 1.2% Other 0.3%



v0.1.2 (Latest)

Optimize the performance

Assets 8

Main program

Asset	SHA256	Size	Time
CrysDiS-Linux-v0.1.2.tar.gz	sha256:f18a57b3f349b884...	226 MB	4 hours ago
CrysDiS-macOS-arm64-v0.1.2.zip	sha256:fdc9186927490382...	112 MB	5 hours ago
CrysDiS-Windows-v0.1.2.zip	sha256:c5a6873b5fd9662a...	139 MB	4 hours ago
CrysDiS_tkinter-Linux-v0.1.2.tar.gz	sha256:29a0f71d932ea181...	193 MB	4 hours ago
CrysDiS_tkinter-macOS-arm64-v0.1.2.zip	sha256:1f44981caeee3c49...	83 MB	4 hours ago
CrysDiS_tkinter-Windows-v0.1.2.zip	sha256:5c0a34cd7721cb37...	111 MB	4 hours ago

Source code

Asset	Time
Source code (zip)	5 hours ago
Source code (tar.gz)	5 hours ago

Tkinter versions are also provided if you prefer a more nostalgic interface

- Please refer to the README file if you need more information

Start the program

Option 3: Run the source code from GitHub

- First install Git and Anaconda/Miniconda. Then clone the repository:

```
git clone https://github.com/hepeng1024/CrysDiS.git
```

```
cd CrysDiS
```

- Create the conda environment:

```
conda env create -f environment.yml
```

```
conda activate crysdis
```

- Start CrysDiS:

```
python CrysDiS.py
```

- Then open this address in a browser:

```
http://127.0.0.1:8080
```

- If port 8080 is already being used, choose another port:

```
NICEGUI_PORT=8094 python CrysDiS.py
```

- Please refer to the README file at <https://github.com/hepeng1024/CrysDiS> for more details

Starting interface

CrysDiS ? INTRO ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar ADVANCED CRYSTAL LIST LOAD CIF CRYSTAL BUILDER ADD COMBO PANEL ADD PANEL

1 Crystal FCC Zone axis 100 Plane Vector Rotation APPLY SYNC



The interface displays two main panels. The left panel shows a 3D crystal structure of FCC (Face-Centered Cubic) with a scale bar of 0.5 nm. The right panel shows a 2D diffraction pattern with a scale bar of 5 nm⁻¹. The diffraction pattern consists of a grid of spots, with the central spot being the largest and most prominent. The spots are arranged in a square grid, with the central spot at the origin. The scale bar in the diffraction pattern is labeled 5 nm⁻¹.

Features of the top toolbar

“ADD PANEL” allows you to have multiple structures at the same time

CrysDIS [?](#) [INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) [CRYSTAL BUILDER](#) [ADD COMBO PANEL](#) **ADD PANEL**

1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)

0.5 nm 5 nm^{-1}

2 Crystal: BCC Zone axis: 110 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)

0.2 nm 5 nm^{-1}

3 Crystal: HCP Zone axis: 0001 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)

0.5 nm 5 nm^{-1}

Features of the top toolbar

“ADD COMBO PANEL” allows you to overlap diffraction patterns from different panels and compare them

CrysDis [INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) [CRYSTAL BUILDER](#) **ADD COMBO PANEL** [ADD PANEL](#)

Panel 1: Crystal: FCC, Zone axis: 100, Plane, Vector, Rotation, APPLY, SYNC. Scale bar: 0.2 nm. Diffraction pattern scale: 5 nm⁻¹.

Panel 2: Crystal: BCC, Zone axis: 110, Plane, Vector, Rotation, APPLY, SYNC. Scale bar: 0.2 nm. Diffraction pattern scale: 5 nm⁻¹.

Panel 3: Crystal: HCP, Zone axis: 0001, Plane, Vector, Rotation, APPLY, SYNC. Scale bar: 0.5 nm. Diffraction pattern scale: 5 nm⁻¹.

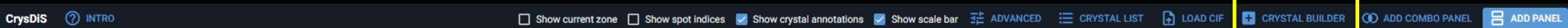
Panel C1: Current crystals: 1. FCC. ADD, REMOVE, Bind crystal motion. Table:

Panel	Crystal	Zone	
1	FCC	[100]	↑ ↓ ×
2	BCC	[110]	↑ ↓ ×
3	HCP	[0001]	↑ ↓ ×

Diffraction pattern scale: 5 nm⁻¹.

Features of the top toolbar

“CRYSTAL BUILDER” lets you build your own crystal



Customized crystal builder

Name
Customized crystal

Lattice
cubic

Symmetry
1: P1

Save target
Project local library

a
0.35 nm

b
0.35 nm

c
0.35 nm

alpha
90 deg

beta
90 deg

gamma
90 deg

APPLY LATTICE CONSTRAINTS

ADD ATOM SITE

PREVIEW SYMMETRY

Atom sites

Atom
Ni

x
0

y
0

z
0

Occ.
1

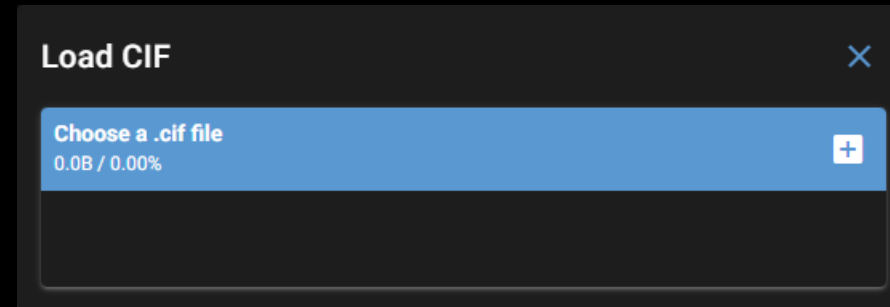
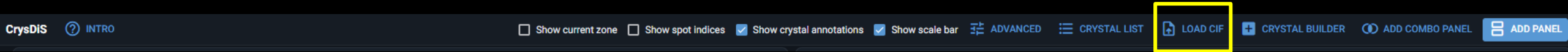
Color
#10B7A5

Site
Ni

SAVE AS A NEW STRUCTURE

Features of the top toolbar

You can load your own .cif files via “LOAD CIF”



Features of the top toolbar

Manage the crystal structures via “CRYSTAL LIST”

CrysDiS ? INTRO

☐ Show current zone

☐ Show spot indices

☒ Show crystal annotations

☒ Show scale bar

⚙ ADVANCED

☰ CRYSTAL LIST

📄 LOAD CIF

⊕ CRYSTAL BUILDER

🔗 ADD COMBO PANEL

📄 ADD PANEL

Three built-in structures that cannot be edited

Crystal list

● FCC	Built-in · cubic, Fm-3m, 1 site(s)	✎	🗑
● BCC	Built-in · cubic, Im-3m, 1 site(s)	✎	🗑
● HCP	Built-in · hexagonal, P1, 2 site(s)	✎	🗑
● (Al,Ti)Ni ₃	Custom · cubic, P1, 5 site(s)	✎	🗑
● Al (mp-134)	Custom · cubic, Fm-3m, 1 site(s)	✎	🗑
● AlNi ₃ (mp-2593)	Custom · cubic, Pm-3m, 2 site(s)	✎	🗑
● D024 Ni ₃ Ti	Custom · hexagonal, P6 ₃ /mmc, 4 site(s)	✎	🗑
● L10 MnNi	Custom · tetragonal, P4/mmm, 3 site(s)	✎	🗑
● L12 Ni ₃ Ti	Custom · cubic, Pm-3m, 2 site(s)	✎	🗑
● Mg (mp-153)	Custom · hexagonal, P6 ₃ /mmc, 2 site(s)	✎	🗑
● Mg ₅ Si ₆ (mp-568306)	Custom · monoclinic, C2/m, 6 site(s)	✎	🗑
● Mg ₅ Si ₆ _no_sym (mp-568306)	Custom · triclinic, P1, 11 site(s)	✎	🗑
● Ni ₃ Si (mn-828)		✎	🗑

Features of the top toolbar

Advanced settings in “ADVANCED”

CrysDiS [INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar **ADVANCED** [CRYSTAL LIST](#) [LOAD CIF](#) [CRYSTAL BUILDER](#) [ADD COMBO PANEL](#) [ADD PANEL](#)

Advanced settings

Simulation method
Pymatgen TEMCalculator

Camera length
200 mm

Voltage
500 kV

Thickness
10 nm

Max hkl
7

Spot intensity threshold
0.001

Compression factor
100
Contrast of diffraction spots intensity

Spot size scaling
0.3
How large bright spots are

☒ Always snap back view
Check to rotate the crystal back when moving to a new zone

☒ Use four-index basis for hexagonal systems

☐ Auto sync crystal/diffraction

PALETTE FOR VECTORS/PLANES

Log
"1 123" is not a valid index for FCC
Panel 1: snapped back to [100]
"1- 123" is not a valid index for FCC
Panel 1: snapped back to [100]
Panel 1: snapped back to [100]

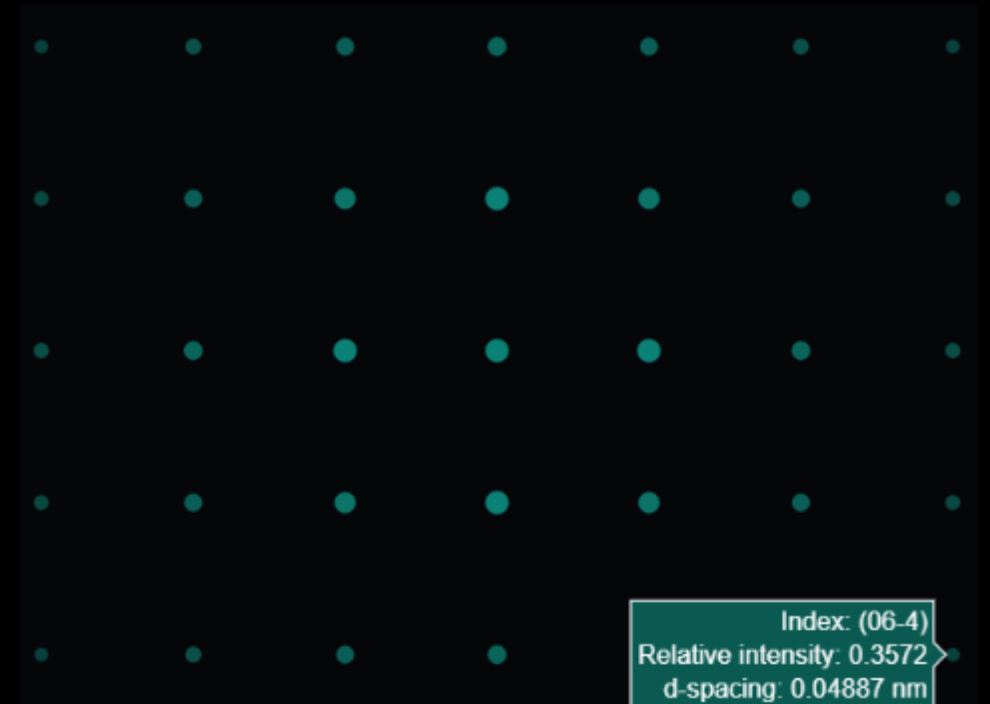
More about advanced settings

- Compression factor

Compression factor = 10



Compression factor = 150



Dim spots are more visible

More about advanced settings

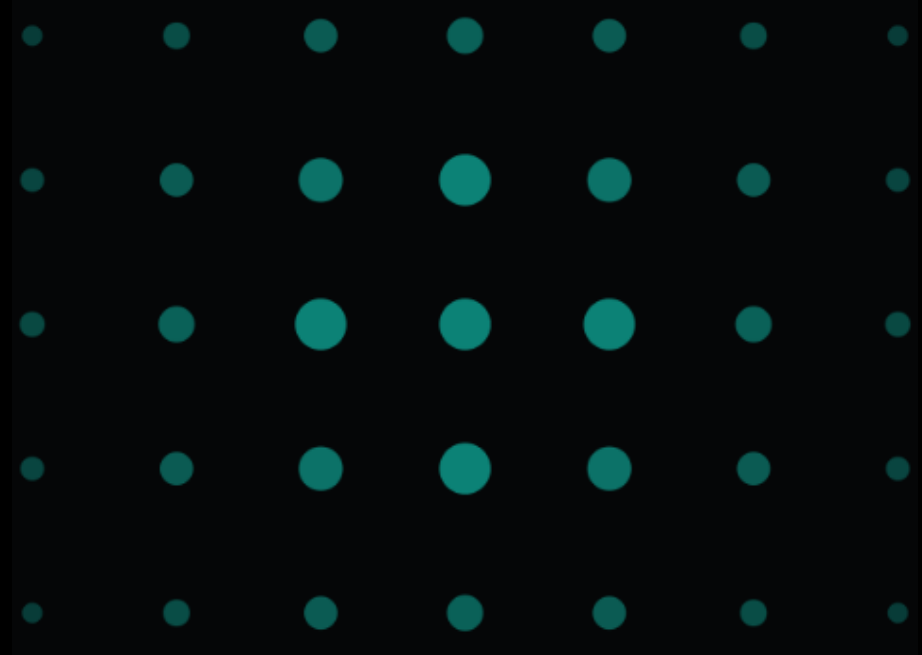
- Spot size scaling effect

Scaling factor = 0



All spots are the same size

Scaling factor = 1



Spot size scales significantly with intensity

More about advanced settings

- Always snap back view



In-plane rotation by 45 deg



Go to the 110 zone

“Always snap back view” is checked



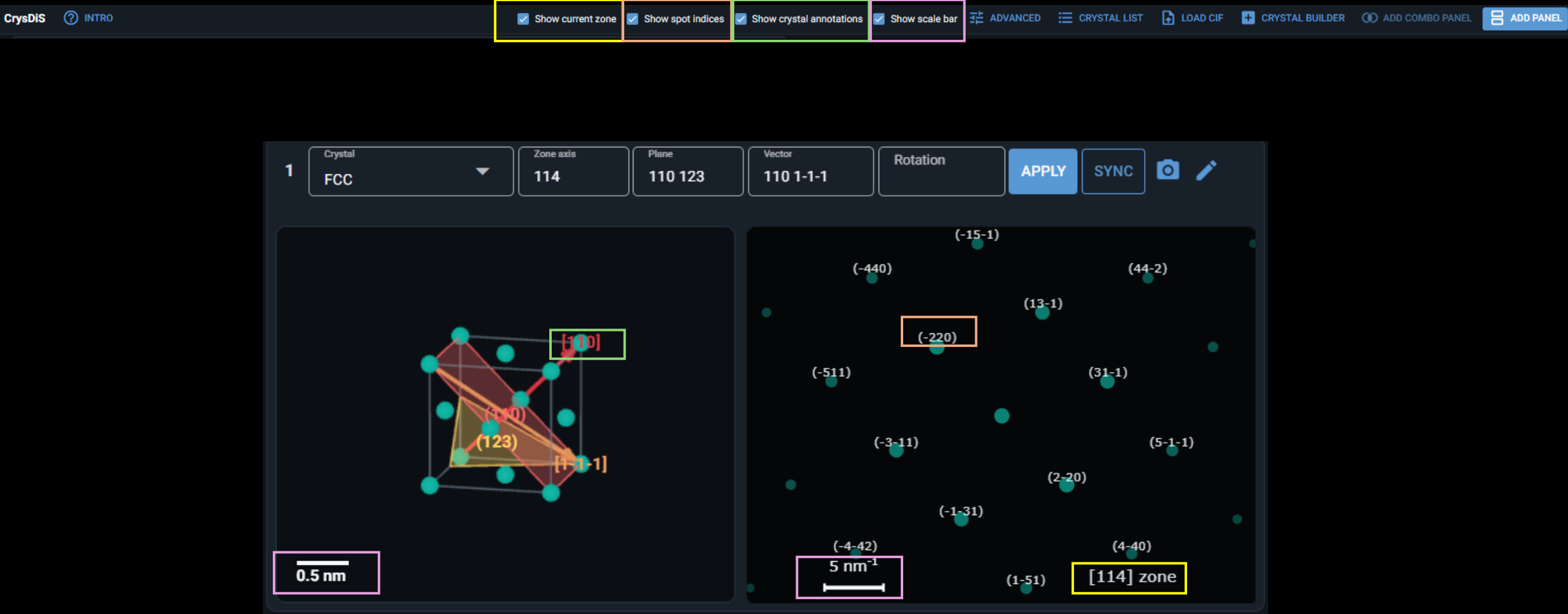
“Always snap back view” is unchecked



The in-plane 45 deg rotation history is retained

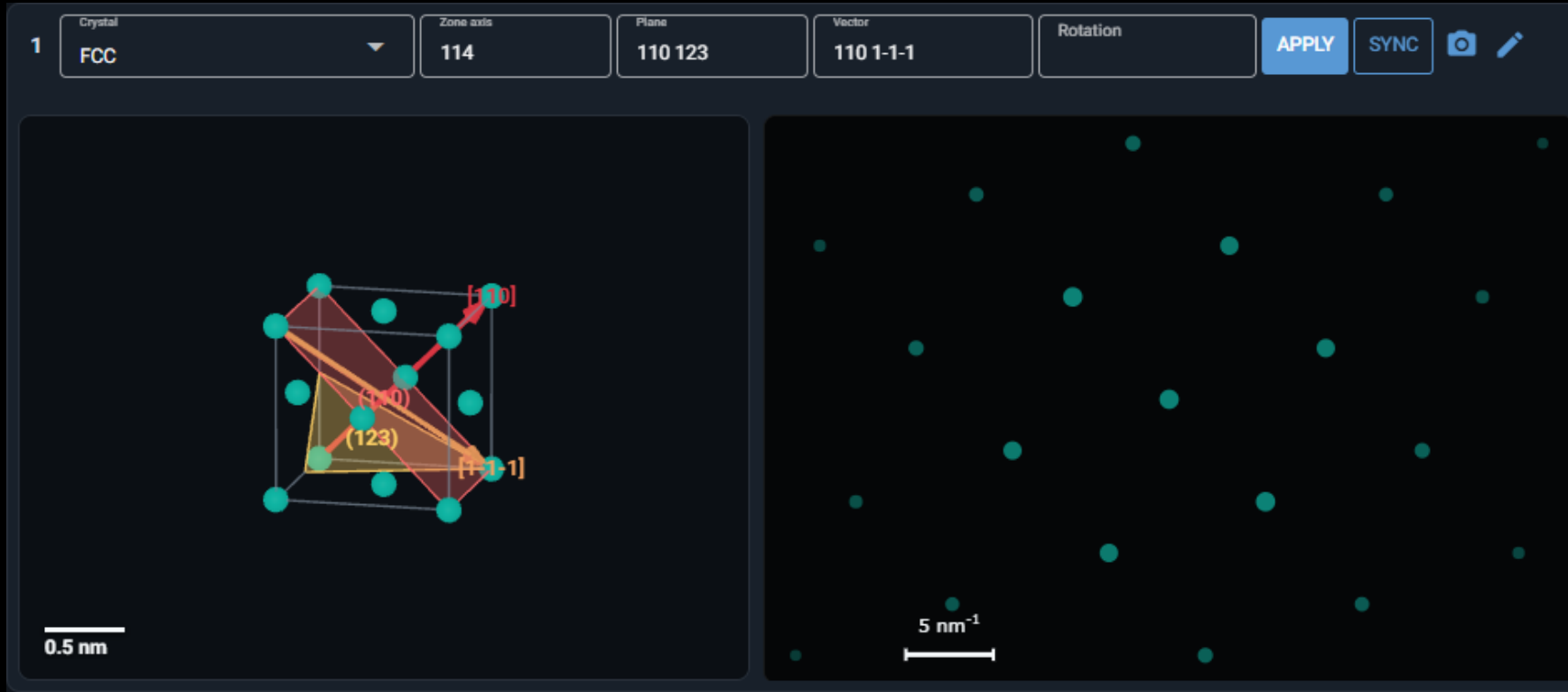
Features of the top toolbar

Some annotation toggles



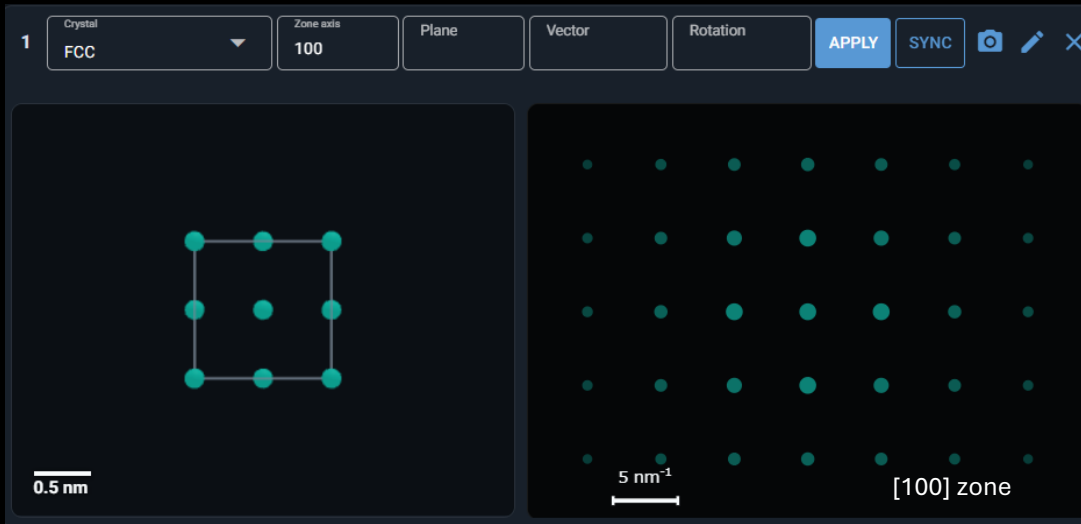
Features of the panel toolbar

- Choose any crystals in the list or upload/build a new one
- You can go to any zone axis, create vectors/planes
- Reciprocal lattice vectors can be created using *hkl (ex. *110, *0-10)
- For hexagonal systems, 4-index basis inputs are allowed (ex. 2-1-10, *0001)

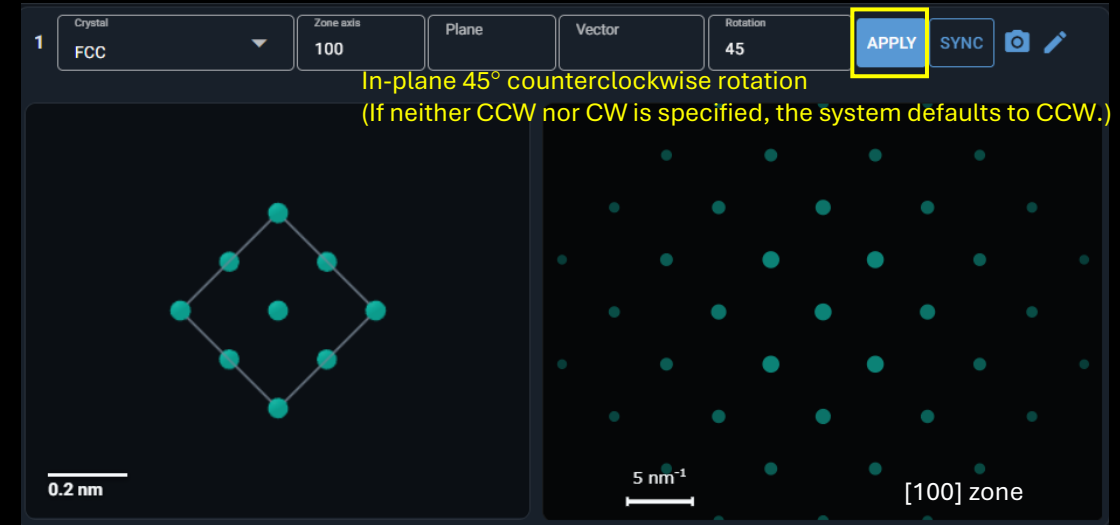


Features of the panel toolbar

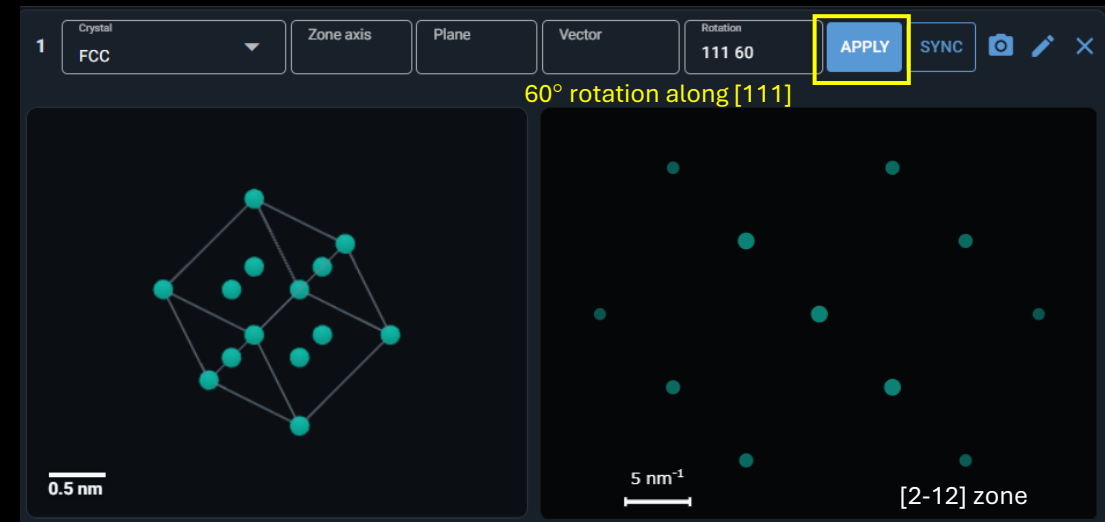
- You can also rotate the crystal along a certain axis
- Right-hand rule is applied



In-plane rotation: use “deg”, “deg cw”, or “deg ccw”

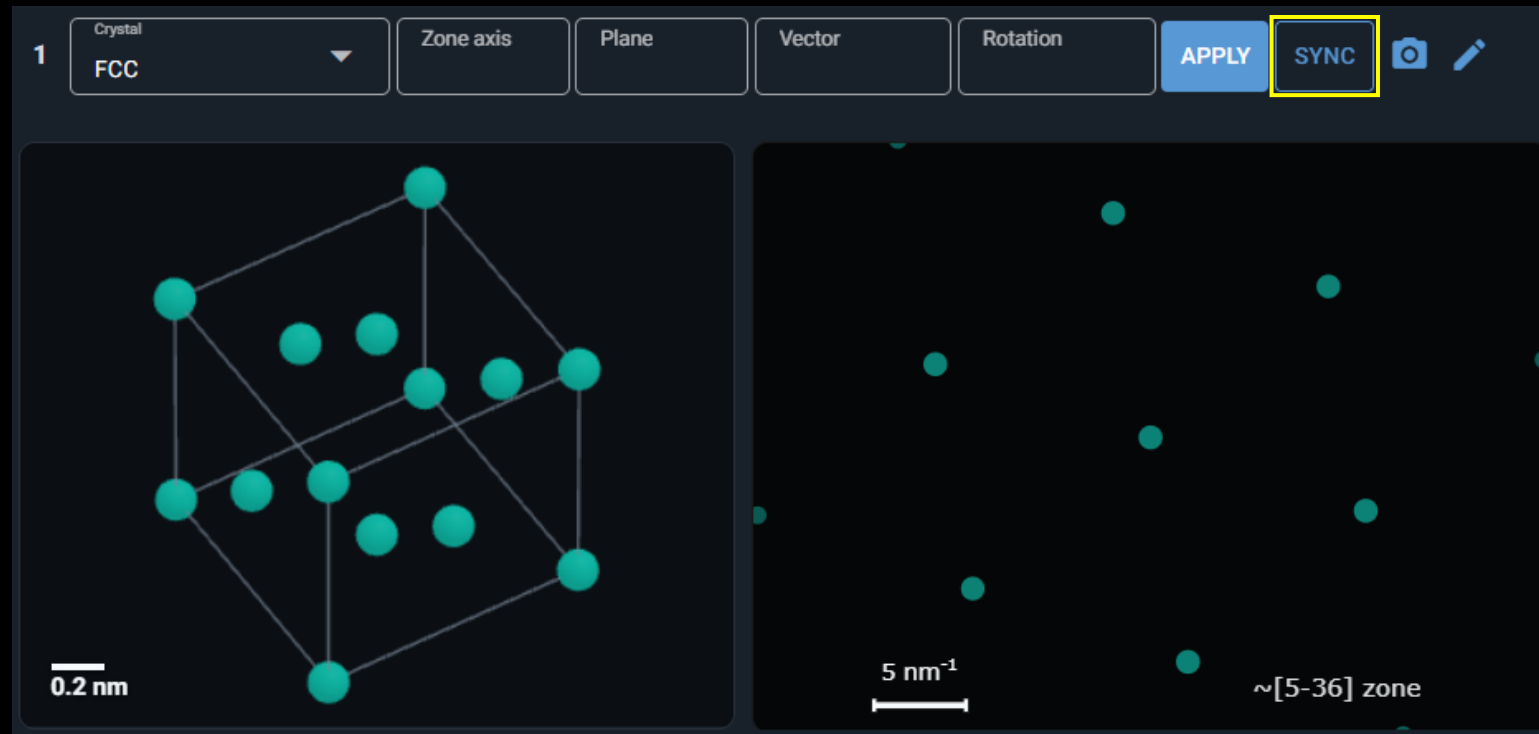


Rotation along a certain axis: use “vector deg”



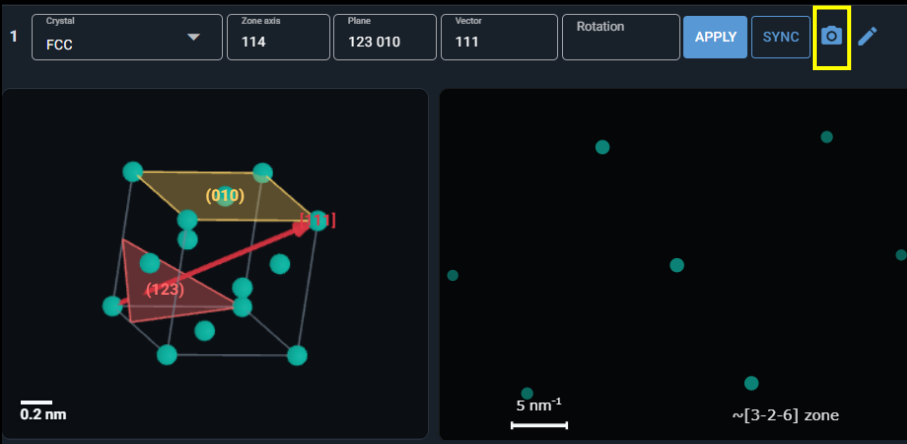
Features of the panel toolbar

- Synchronize the crystal and diffraction pattern using “SYNC”
- It will detect the closest zone axis
- You can also turn on the “Auto sync crystal/diffraction” in the advanced settings



Features of the panel toolbar

- Download the crystal/diffraction pattern at the current viewing angle



Download panel 1

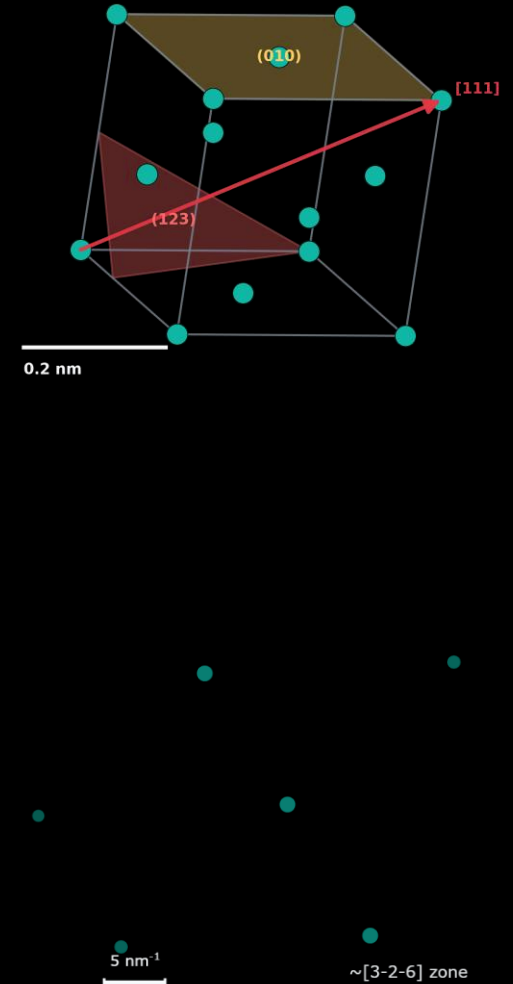
☒ Real-space crystal

☒ Diffraction pattern

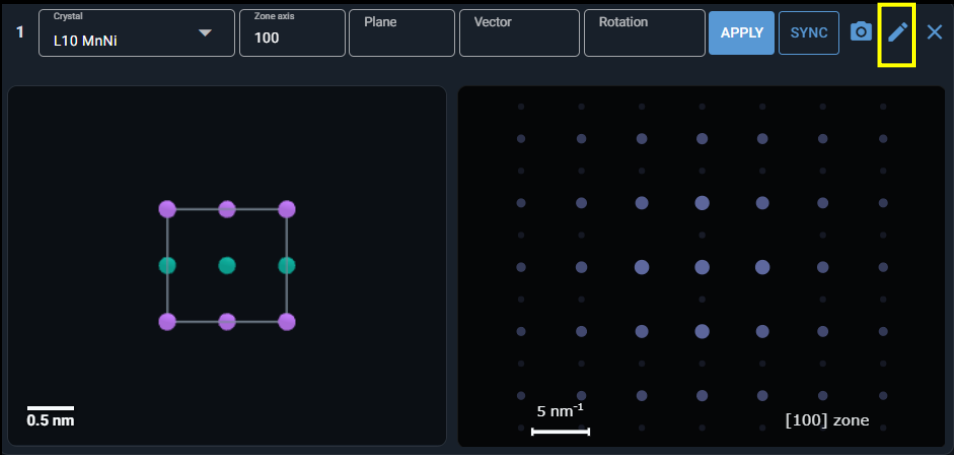
☒ Transparent background

Quality
300 dpi

CANCEL [DOWNLOAD]



Features of the panel toolbar



- Edit the current structure

Customized crystal builder

Name

L10 MnNi

Lattice

tetragonal

Symmetry

123: P4/mmm

Save target

Project local library

a

0.374

nm

b

0.374

nm

c

0.352

nm

alpha

90

deg

beta

90

deg

gamma

90.0

deg

APPLY LATTICE CONSTRAINTS

+

ADD ATOM SITE

PREVIEW SYMMETRY

Atom sites

Atom

Mn

x

0

y

0

z

0

Occ.

1

Color

#C77DFF

Site

Mn1

Atom

Mn

x

0.5

y

0.5

z

0

Occ.

1

Color

#C77DFF

Site

Mn2

Atom

Ni

x

0.5

y

0

z

0.5

Occ.

1

Color

#10B7A5

Site

Ni1

SAVE EDITED STRUCTURE

SAVE AS A NEW STRUCTURE

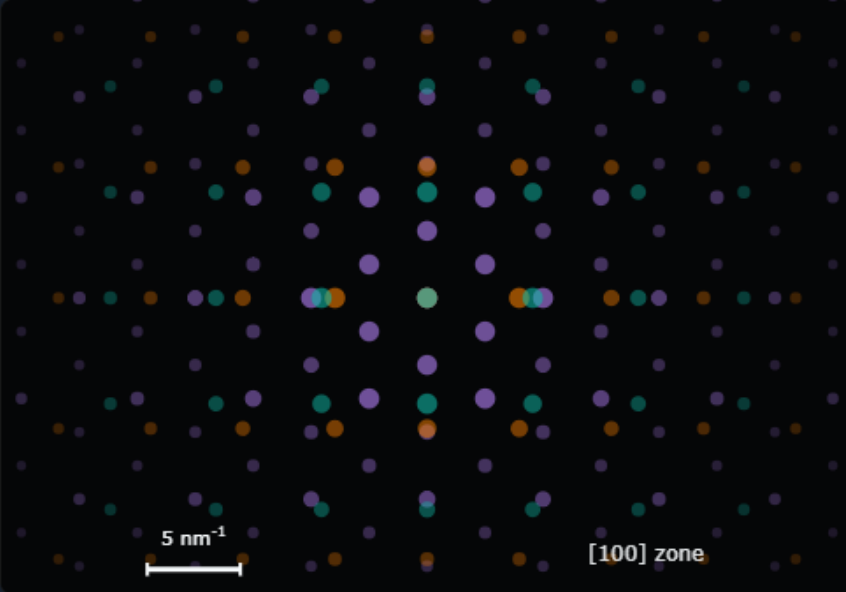
Features of the combo panel

Select crystals present on the page

Add or remove it

C1 Current crystals 1. FCC ADD REMOVE ☐ Bind crystal motion  ×

Panel	Crystal	Zone	
1	FCC	[100]	↑ ↓ ×
2	BCC	[110]	↑ ↓ ×
3	HCP	[0001]	↑ ↓ ×



5 nm⁻¹ [100] zone

The interface shows a list of crystals and their corresponding diffraction patterns. The current crystal is FCC, and the current zone is [100]. The diffraction pattern is a 2D grid of colored dots (purple, green, orange) on a dark background. A scale bar indicates 5 nm⁻¹ and the zone is labeled [100] zone.

Ex. FCC diffraction pattern will be shown on top

The current zone shows the zone axis of the crystal on the top-most layer

The order of the crystal in the list determines the order of the diffraction patterns

Features of the combo panel

Check “Bind crystal motion” if you want the crystals to synchronize their motions

C1

Current crystals
1. FCC

ADDREMOVE☒ Bind crystal motion📷✕

Panel	Crystal	Zone	
1	FCC	[100]	↑ ↓ ✕
2	BCC	[110]	↑ ↓ ✕

5 nm⁻¹

[100] zone

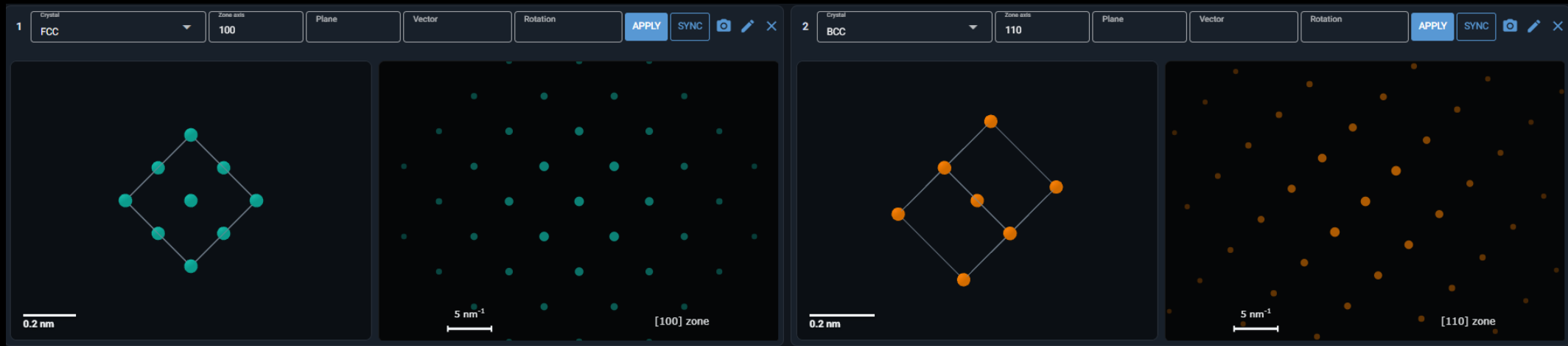
Features of the combo panel

- Bind crystal motion



Ex. if you rotate the crystal in panel 1 by 45° ccw

the crystal in panel 2 will do the same motion



In this case, it means crystals 1 & 2 have the orientation relationship: $[100]_{\text{crystal1}} // [110]_{\text{crystal2}}$

Comparison of two simulation methods

Simulation method	Ewald Sphere Kinematic	Pymatgen TEMCalculator
Main purpose	More TEM-like kinematic spot visibility	Fast crystallographic zone-axis pattern
Uses voltage?	Yes, through electron wavelength and Ewald sphere radius	Only for wavelength calculation
Uses thickness?	Yes	No
Includes relrod / excitation error?	Yes	No
Includes Ewald sphere curvature?	Yes	No
Good for	Showing why spots appear/disappear with tilt/thickness	Clean ideal zone-axis indexing/intensity